

into the early hours of the morning. Overclocking only began after that. The first trial run showed us that we could cross 4GHz without any issues. This was only in the BIOS. We would have no issues with posting. The BIOS was setup to disable any thermal protection and alerts. There would be no throttling down of the CPU, so we would know immediately if we were going to face instability. Throughout our exercise we noticed BSODs during bootup, and sometimes while running benchmarks.

We would use CPU-Z, GPU-Z and SpeedFan to monitor the various statistics of the graphics card and processor. Cinebench was our stress test for the processor once we over-clocked it a step higher. A 100 per cent load on the CPU for more than 30 seconds would result in an instant bluescreen of death or a lockup.

Finally, we started our overclocking run. Initially, the clocks were pushed higher by simply switching multipliers and we were easily able to touch around 4.1GHz without much effort. Once we reached 4.2 GHz or so, we hit a road-



Attaching the mount for the Corsair H50 cooler

block. Depending more on the board's clock frequency was the way to go. We cut down on the multiplier and started focussing on incrementing the FSB. Finally around 4.3GHz, we hit another



The two Radeon 5970s ready for action

roadblock. The overclock was stable till we booted into Windows. Benchmarks would throw up BSODs and sometimes even just a CPU-Z or SpeedFan application would cause instability.

It was time to tweak the voltages to improve stability. We went with over-volting the processor by a bit to get that stability. After multiple combinations, tweaks and reboots, we hit our magical figure of 4.4 GHz or 4412 MHz to be precise. We were able to cross 4.5GHz and boot into Windows, but not enough to achieve stability for benchmarks.

The Results

The results of the overclock and build are nothing short of impressive, even for us. Majority of the overclockers online don't cross 4.1 GHz overclocks on a 980X. The 4.4 GHz overclock we achieved, lets the processor idle at around 40 C and under load, it peaks out at around 63 C. We can assume this setup to be stable enough to run day and night without any problems.

The Corsair H50 does a great job of cooling the Core i7 and although compact and simplistic, it isn't far behind large heavy duty cooling units.

Coming to scores, the stock setup itself was the fastest we had seen and the likelihood of anyone using such a system is pretty slim. The stock frame rates in Far Cry 2 at the absolutely highest quality settings, was mind boggling. Turning on FSAA to 4x or 8x hardly made a difference to the scores.

Overclocking only boosted this performance by a lot. The overclocked processor ran 32 per cent faster. For the fastest desktop processor that Intel sells, it's a lot. The performance wasn't only limited to processing and graphics rendering either. The RAID setup drives were churning out numbers not reachable by standard drives. To give you a gist, we were seeing 200 MB/s read speeds. The graphics cards we used were extremely hot. Fortunately, they were factory overclocked but we would have liked to push them a little bit further. It would have helped us reach the 44,000 figure in 3DMark Vantage.

What we've managed to do is build the fastest gaming PC in the country. It doesn't look like this kind of performance will be available on a single graphics card for at least a year to come. The approximate cost of the entire system was around 2.5 lac rupees. We've achieved what we set out to build, and most of the guys at Digit should be occupied with playing games at the most ridiculous quality settings ever. 

CONFIGURATION

Processor	Intel Core i7 980X
Motherboard	ASUS Rampage III Extreme
Graphics Card	XFX Radeon 5970 Black Edition (x2)
Memory	Kingston HyperX DDR3 (6 x 2 GB)
Hard Drives	WD Velociraptor 300 GB (x2)
Power Supply	Cooler Master's Silent Pro 1200W
Chassis	Cooler Master HAF X
Mouse	Razer Lachesis